

Idovonzas Torvenye - Time-Attraction Law

Unified Effective Density Framework - Idokamra 1.0 v8.3 FINAL

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Abstract (EN)

This document provides the final corrected derivation for Time-Attraction Law v8.3. Corrigendum v8.2 fixes $R=(r+1)/(r+2)$ to the correct Bellini & Sawicki (2014) eq 2.18 $R=(1+r)/(1+2r)$ where $r=\alpha_M/\alpha_B$. Update v8.3 derives epsilon in closed form on subclass $G5=0$, $G4=G4(\phi)$: $M^2=2G4$, $\alpha_M=2 \phi \dot{G4} / (H M^2)$, $H M^2 \alpha_B=2 \phi \dot{[X G3 - G4\phi]} \rightarrow r=G4\phi / (X G3 - G4\phi)$, $\epsilon=X G3/G4\phi - 2$. Consequently f_L cancels: $r=1/(1+\epsilon)$, low-density limit independent of screening form. Sign sorted via $dR/dr=-1/(1+2r)^2 < 0$. Background upgrade replaces local ρ_L with background $\rho_m(a)=\rho_{m0} a^{-3}$, so $f_L \rightarrow f_L(a)$, $\alpha_M(a)$, $\alpha_B(a)$ are time functions as required by EFT of dark energy, yielding binned $R(a)$ testable via hi_class against Euclid/DESI. No hand-chosen 1.2 and 2, shape falsifiable. SPARC rotation curves contain no per-galaxy lensing, thus $\Delta\sigma$ not available and R cannot be formed from SPARC; 76-95% accuracy framing removed, replaced by hi_class binned $R(a)$ forecast (template Sec X enhanced-early-gravity cubic Galileon recovers Planck mass within 0.03% across 4 redshifts, 3 wavenumbers). Framework remains falsifiable, dimensionally consistent (kg/m³). Acknowledgment to Jordi Audet Palau for Sec V, VI, X corrections.

1. Corrigendum v8.2 - R formula javitasa

v8.1-ben hibásan $R=(r+1)/(r+2)$ szerepelt. Helyes forma Bellini & Sawicki (2014) Eq 2.18 es Jordi Audet Palau note alapján:
 $R = \Delta\sigma/\Delta\mu = (1+r)/(1+2r)$, $r = \alpha_M/\alpha_B$
Inverz: $r = (1-R)/(2R-1)$
Ellenorzes: $r=1 \rightarrow R=2/3$ mindket kepletben, de $r \neq 1$ eseten elter: $r=0.5$: regi 0.60 $\rightarrow$ helyes 0.75; $r=2$: regi 0.75 $\rightarrow$ helyes 0.60; $r=5$: regi 0.857 $\rightarrow$ helyes 0.545
 $dR/dr = -1/(1+2r)^2 < 0$ monoton csokkeno.

2. Update v8.3 - epsilon zart alak levezetese (Per Jordi Sec V)

Feltetel: Horndeski alosztaly $G5=0$, $G4=G4(\phi) \rightarrow G4X=0$. EFT parametererek:
 $M^2 = 2 G4$
 $\alpha_M = 2 \phi \dot{G4} / (H M^2)$
 $H M^2 \alpha_B = 2 \phi \dot{[X G3 - G4\phi]}$
Osztas: $r = \alpha_M/\alpha_B = G4\phi / (X G3 - G4\phi)$
 $1+\epsilon = \alpha_B/\alpha_M = 1/r \rightarrow \epsilon = 1/r - 1 = X G3/G4\phi - 2$ QED
Locus: $\epsilon=0 \leftrightarrow X G3 = 2 G4\phi \leftrightarrow r=1 \leftrightarrow R=2/3$
 f_L kiesik: $r = \alpha_0 f_L / [\alpha_0 f_L (1+\epsilon)] = 1/(1+\epsilon) \rightarrow$ fuggetlen screening formatol
Elojell: $dR/dr < 0 \rightarrow X G3 > 2 G4\phi \rightarrow \epsilon > 0 \rightarrow r < 1 \rightarrow R$ felulrol; $X G3 < 2 G4\phi \rightarrow \epsilon < 0 \rightarrow r > 1 \rightarrow R$ alulrol (Sec4 ezt keri).

3. Background Correction v8.3

Problema: α_M , α_B csak idofuggo hatterfuggvenyek EFT-ben. Lokalis ρ_L nem parameteruk $\rightarrow f_L(\rho_L)$ nem illesztheto.
Megoldas: ρ_L legyen hatter: $\rho_m(a)=\rho_{m0} a^{-3} \rightarrow f_L \rightarrow f_L(a)$, $\alpha_M(a)$, $\alpha_B(a)$, $\epsilon(a)=X G3/G4\phi - 2$, $r(a)=1/(1+\epsilon(a))$, $R(a)=(1+r(a))/(1+2r(a))$ binned a-ban $\rightarrow hi_class \rightarrow$ Euclid/DESI teszt.
Template Sec X: enhanced-early-gravity cubic Galileon $r=-1.2$ to -1.3 visszahozza Planck

tomeget 0.03%-on.

4. Sec5 SPARC Revision

SPARC = rotacios gorbek + Spitzer fotometria, nincs per-galaxis lensing -> DeltaSigma nincs -> R nem kepezhető SPARC-ból. 76-95% framing torolve. Csere: hi_class binned R(a) forecast Sec VI. Regi 94.9% SPARC 150 -> Uj R(a) binned Euclid/DESI.

5. Unified Framework kg/m³

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rho_eff = rho_b*M_rho + rho_DM*M_rho + rho_B*M_B + rho_GR*M_B + rho_E*M_B + rho_psi
rho_psi = Xi*epsilon0*Phi_t, Phi_t=0.6*EEG_coh+0.4*HRV_coh
v8.3: rho_L->rho_m(a), epsilon(a), r(a), R(a) - dimenzio-helyes, falszifikalhato.
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6. Kepletek egy helyen - Fuggelek

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R = (1+r)/(1+2r)
r = (1-R)/(2R-1) = G4phi/(X G3X - G4phi) = 1/(1+epsilon)
epsilon = X G3X/G4phi -2
dR/dr = -1/(1+2r)^2
M^2=2G4, alpha_M=2 phidot G4phi/(H M^2), H M^2 alpha_B=2 phidot[X G3X-G4phi]
rho_m(a)=rho_m0 a^-3, R(a)=(1+r(a))/(1+2r(a)) binned
Phi_t=0.6*EEG_coh+0.4*HRV_coh
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